

$$\begin{aligned}
& \frac{-1}{16\pi^2} \left(-\frac{2}{81} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{108} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{8}{405} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_d^{\mathbf{P}}] \right. \\
& \quad \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{36} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{8}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} \sum_{\mathbf{l}} g_1^4 LF_{3,0}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{5}{72} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{81} \sum_{\mathbf{q}} g_1^4 LF_{3,0}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{216} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{405} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{8}{81} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{5}{27} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{32}{405} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{27} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{18} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{135} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{36} g_1^4 LF_{2,1,0}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 LF_{2,2,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{18} g_1^4 LF_{3,1,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 LF_{4,1,-2}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{36} g_1^4 LF_{2,1,0}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 LF_{2,2,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{18} g_1^4 LF_{3,1,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 LF_{4,1,-2}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{1296} g_1^4 LF_{2,1,0}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{2,2,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{648} g_1^4 LF_{3,1,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{4,1,-2}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{1296} g_1^4 LF_{2,1,0}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{2,2,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{648} g_1^4 LF_{3,1,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{4,1,-2}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{48} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{24} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{48} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{24} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{9} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{2,1,0}[m_d^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{2,2,-1}[m_d^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} + \\
& \quad \frac{1}{18} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{3,1,-1}[m_d^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} + \frac{1}{18} g_1^4 LF_{2,1,0}[m_e^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{36} g_1^4 LF_{3,1,-1}[m_e^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^4 LF_{2,1,0}[m_e^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{36} g_1^4 LF_{3,1,-1}[m_e^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{2,1,0}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} - \\
& \quad \frac{1}{36} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{2,2,-1}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{36} g_1^2 \overline{y}_e^{pi3} y_e^{pi4} LF_{3,1,-1}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} + \\
& \quad \frac{1}{648} g_1^4 LF_{2,1,0}[m_q^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{3,1,-1}[m_q^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{24} g_1^2 g_2^2 LF_{2,1,0}[m_q^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{3,1,-1}[m_q^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_q^{i1}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_q^{i1}, m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{648} g_1^4 LF_{2,1,0}[m_q^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{3,1,-1}[m_q^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{24} g_1^2 g_2^2 LF_{2,1,0}[m_q^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{3,1,-1}[m_q^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_q^{i2}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_q^{i2}, m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{9} g_1^2 \overline{y}_u^{i2p} y_u^{i1p} LF_{2,1,0}[m_u^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} - \frac{1}{9} g_$$